



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 7

1. State the Biot-Savart's law and then derive an expression for the magnetic field intensity $\mathbf{H}$ at the field point P due to a straight current-carrying filamentary conductor of finite length AB?
2. Find the magnetic field intensity $\mathbf{H}$ at the field point P (-3, 4, 0) due to current-carrying filamentary conductor as shown in Fig. (1).

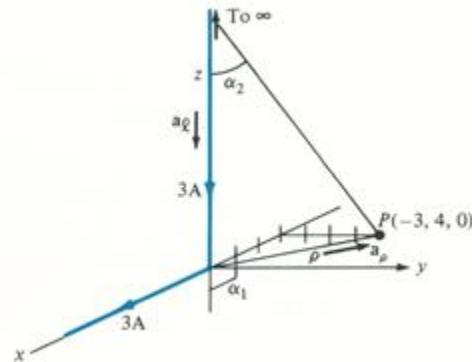


Fig. (1). A current filament along the semi-infinite x- and z- axes.

3. Derive an expression for the magnetic field $\mathbf{H}$ at a point P (0, 0, h) due to a circular loop of radius a and carrying a current I. Then find $\mathbf{H}$ at (0, 0, 1) cm if $a=5\text{ cm}$ and $I=50\text{ mA}$ if the ring is placed at $z=1$ plane?